Assignment 6

Subject - PSDL - II
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Batch - B3

Problem Statement :

Develop a multimedia application using Flutter that demonstrates audio and video playback functionalities.

Code :

main.dart

```
import 'package:flutter/material.dart';
import 'package:audioplayers/audioplayers.dart';
import 'package:video_player/video_player.dart';

void main() {
 runApp(MyApp());
}

class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
   return MaterialApp(
     debugShowCheckedModeBanner: false,
     theme: ThemeData(primarySwatch: Colors.deepPurple),
     home: MultimediaScreen(),
   );
 }
}
```

```
class MultimediaScreen extends StatefulWidget {
 @override
 _MultimediaScreenState createState() => _MultimediaScreenState();
}

class _MultimediaScreenState extends State<MultimediaScreen> {
 final AudioPlayer audioPlayer = AudioPlayer();
 late VideoPlayerController videoController;

 bool isPlayingAudio = false;

 Duration audioPosition = Duration.zero;
 Duration audioDuration = Duration.zero;

 @override
 void initState() {
   super.initState();

   // VIDEO INIT
   videoController = VideoPlayerController.asset('assets/video.mp4')
     ..initialize().then((_) {
       setState(() {});
     });

   // AUDIO LISTENERS (FIXED)
   audioPlayer.onPositionChanged.listen((pos) {
     if (mounted) {
       setState(() => audioPosition = pos);
     }
   });

   audioPlayer.onDurationChanged.listen((dur) {
     if (mounted) {
       setState(() => audioDuration = dur);
     }
   });
 }

 @override
 void dispose() {
   audioPlayer.dispose();
   videoController.dispose();
   super.dispose();
 }

 // AUDIO CONTROLS
 void toggleAudio() async {
   if (isPlayingAudio) {
     await audioPlayer.pause();
```

```
  } else {
    await audioPlayer.play(AssetSource('audio.mp3'));
  }
  setState(() => isPlayingAudio = !isPlayingAudio);
}

void stopAudio() async {
  await audioPlayer.stop();
  setState(() {
    isPlayingAudio = false;
    audioPosition = Duration.zero;
  });
}

void seekAudio(double value) async {
  await audioPlayer.seek(Duration(seconds: value.toInt()));
}

// VIDEO CONTROLS
void toggleVideo() {
  if (videoController.value.isPlaying) {
    videoController.pause();
  } else {
    videoController.play();
  }
}

void seekVideo(double value) {
  videoController.seekTo(Duration(seconds: value.toInt()));
}

void forward10(Duration position) {
  videoController.seekTo(position + Duration(seconds: 10));
}

void backward10(Duration position) {
  videoController.seekTo(position - Duration(seconds: 10));
}

String format(Duration d) {
  String two(int n) => n.toString().padLeft(2, '0');
  return "${two(d.inMinutes)}:${two(d.inSeconds % 60)}";
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: Text("Multimedia Player"),
```

```
      centerTitle: true,
    ),
    body: Container(
      padding: EdgeInsets.all(16),
      decoration: BoxDecoration(
        gradient: LinearGradient(
          colors: [Colors.deepPurple.shade100, Colors.white],
          begin: Alignment.topCenter,
          end: Alignment.bottomCenter,
        ),
      ),
      child: Column(
        children: [

          // AUDIO CARD
          Card(
            shape: RoundedRectangleBorder(
                borderRadius: BorderRadius.circular(20)),
            elevation: 6,
            child: Padding(
              padding: const EdgeInsets.all(16),
              child: Column(
                children: [
                  Text("Audio Player",
                      style: TextStyle(
                          fontSize: 20, fontWeight: FontWeight.bold)),

                  Icon(Icons.music_note, size: 60),

                  Slider(
                    min: 0,
                    max: audioDuration.inSeconds == 0
                        ? 1
                        : audioDuration.inSeconds.toDouble(),
                    value: audioPosition.inSeconds
                        .clamp(0, audioDuration.inSeconds)
                        .toDouble(),
                    onChanged: seekAudio,
                  ),

                  Row(
                    mainAxisAlignment: MainAxisAlignment.spaceBetween,
                    children: [
                      Text(format(audioPosition)),
                      Text(format(audioDuration)),
                    ],
                  ),

                  Row(
```

```
                mainAxisAlignment: MainAxisAlignment.center,
                children: [
                  ElevatedButton.icon(
                    onPressed: toggleAudio,
                    icon: Icon(isPlayingAudio
                        ? Icons.pause
                         : Icons.play_arrow),
                    label:
                    Text(isPlayingAudio ? "Pause" : "Play"),
                  ),
                  SizedBox(width: 10),
                  ElevatedButton.icon(
                    onPressed: stopAudio,
                    icon: Icon(Icons.stop),
                    label: Text("Stop"),
                    style: ElevatedButton.styleFrom(
                        backgroundColor: Colors.red),
                  ),
                ],
              )
            ],
          ),
        ),
      ),

      SizedBox(height: 20),

      // VIDEO CARD
      Expanded(
        child: Card(
          shape: RoundedRectangleBorder(
              borderRadius: BorderRadius.circular(20)),
          elevation: 6,
          child: Padding(
            padding: const EdgeInsets.all(16),
            child: Column(
              children: [
                Text("Video Player",
                    style: TextStyle(
                        fontSize: 20, fontWeight: FontWeight.bold)),

                videoController.value.isInitialized
                    ? ValueListenableBuilder(
                  valueListenable: videoController,
                  builder: (context,
                      VideoPlayerValue value, child) {
                    final position = value.position;
                    final duration = value.duration;
```

```
                    return Column(
                      children: [
                        AspectRatio(
                          aspectRatio: value.aspectRatio,
                          child:
                          VideoPlayer(videoController),
                        ),

                        Slider(
                          min: 0,
                          max: duration.inSeconds == 0
                              ? 1
                              : duration.inSeconds
                              .toDouble(),
                          value: position.inSeconds
                              .clamp(
                              0, duration.inSeconds)
                              .toDouble(),
                          onChanged: (val) {
                            seekVideo(val);
                          },
                        ),

                        Row(
                          mainAxisAlignment:
                          MainAxisAlignment.spaceBetween,
                          children: [
                            Text(format(position)),
                            Text(format(duration)),
                          ],
                        ),

                        Row(
                          mainAxisAlignment:
                          MainAxisAlignment.center,
                          children: [
                            IconButton(
                              icon: Icon(Icons.replay_10),
                              onPressed: () =>
                                  backward10(position),
                            ),
                            IconButton(
                              icon: Icon(
                                value.isPlaying
                                    ? Icons.pause_circle
                                     : Icons.play_circle,
                              ),
                              iconSize: 40,
                              onPressed: toggleVideo,
```

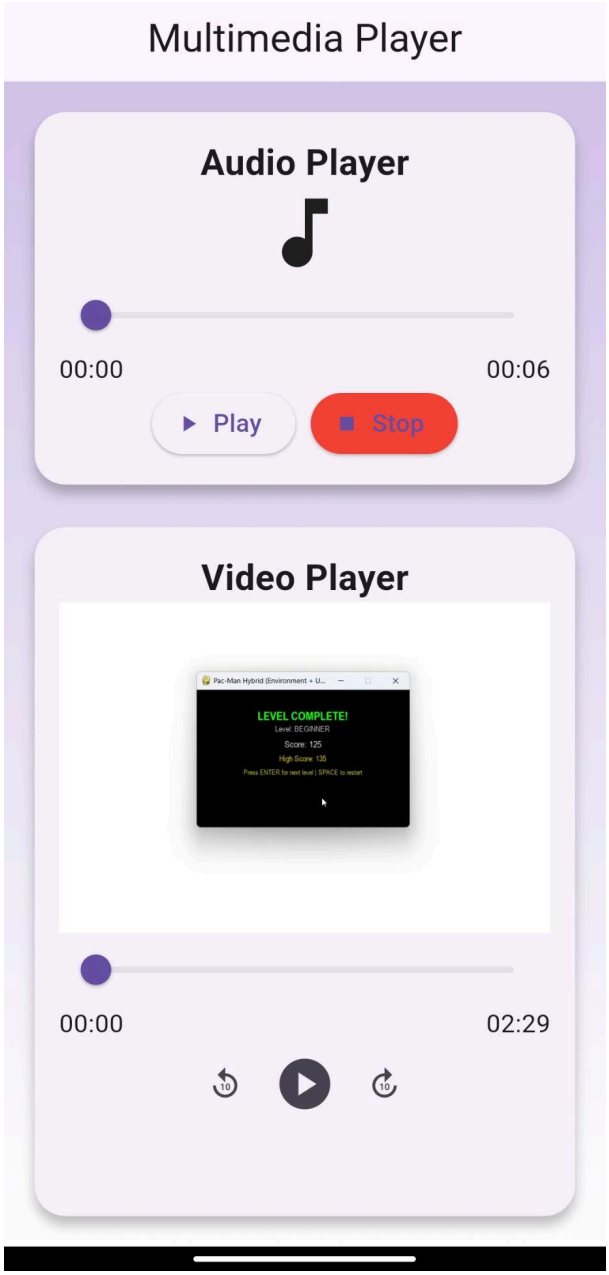
```
                                  ),
                                  IconButton(
                                    icon:
                                    Icon(Icons.forward_10),
                                    onPressed: () =>
                                        forward10(position),
                                  ),
                                ],
                              ),
                            ],
                          );
                        },
                      )
                          : CircularProgressIndicator(),
                    ],
                  ),
                ),
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

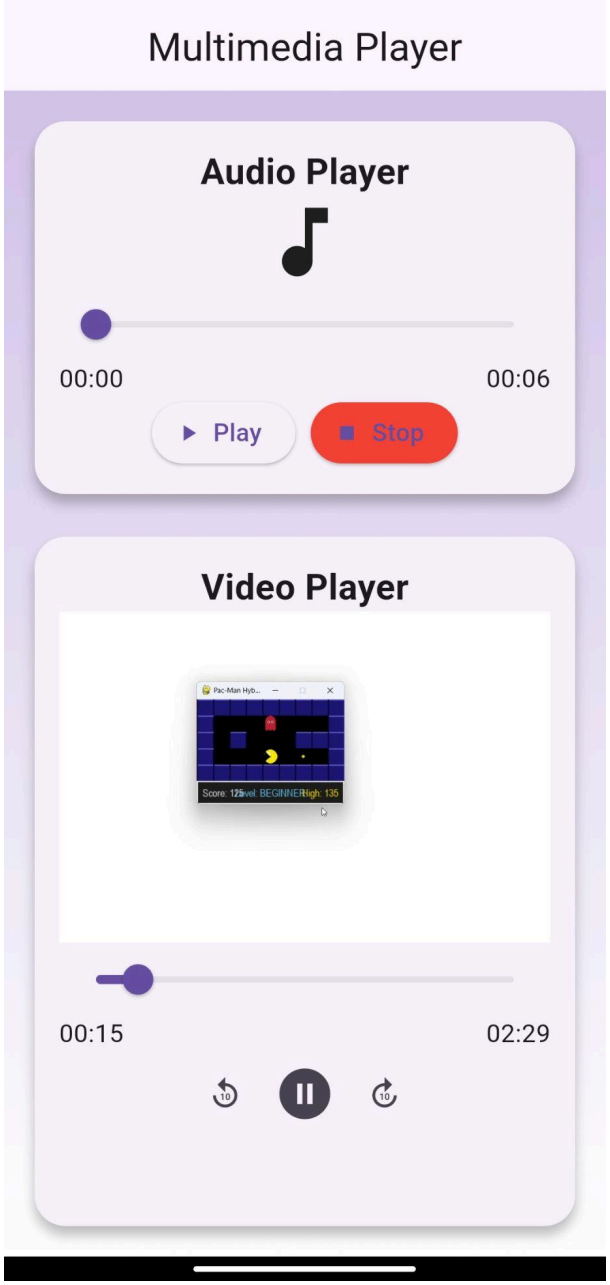
In pubspec.yaml

flutter

```
# The following line ensures that the Material Icons font is
# included with your application, so that you can use the icons in
# the material Icons class.
uses-material-design: true
assets:
  - assets/audio.mp3
  - assets/video.mp4
```

Output :


Multimedia Player
Audio Player
00:00
00:06
Play
Stop
Video Player
LEVEL COMPLETE!
Level: BEGINNER
Score: 125
High Score: 135
Press ENTER for next level | SPACE to restart
00:00
02:29


Multimedia Player
Audio Player
00:00
00:06
Play
Stop
Video Player
00:15
02:29